

Name	Role
Ankur	Presenter, Application, Outcomes
Hasna	Methods and Materials, Negative Outcome
Jessie	Presenter, Positive Outcomes, Methods to Measure Outcomes
Matthew	Presenter, Application, Justification, Snapgene, Cover Slide
Sinyuan (Cindy)	Application, Gene Characteristics
Winston	Presenter, Application, Methods and Materials

Genetic Engineering of Bacteria to Improve Human Health Microbiome Immunology

Group 1 - YaasTea

Ankur, Hasna, Jessie, Matthew, Sinyuan, Winston

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Probiotic Neoantigen Delivery Vectors – Immunotherapy

Application: This method of bacteria genetic engineering uses probiotic **E. coli Nissle 1917 (EcN)**, a non-pathogenic *Escherichia coli* strain.

- Engineered probiotic EcN can be used for **targeting tumours** [1,2].

Justification:

- Tumour neoantigens are present only in specific areas, **minimising risk for autoimmunity** in cancer treatment [3].
- Bacteria are able to **stimulate the immune system** due to their foreign properties [4].
- This technology would be more efficient than traditional neoantigen vaccines due to their **long lasting effectivity**.

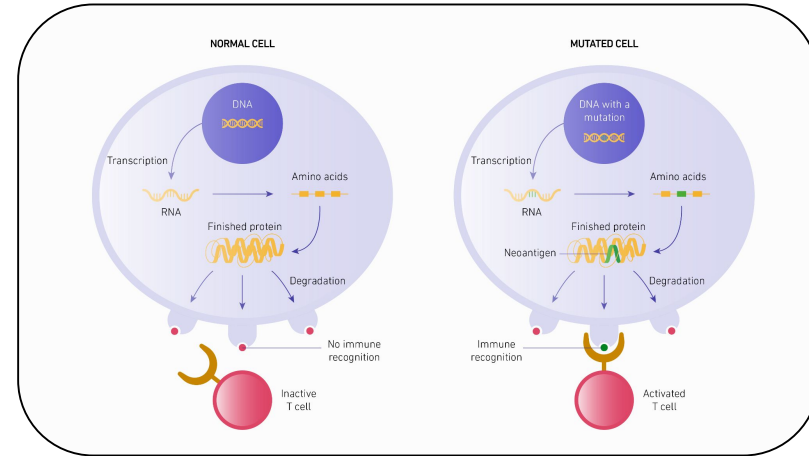


Figure 1: Differences in antigen presentation in a normal cell (left) compared to **neoantigen production and presentation** in a mutated cell (right). Credit: Technology Networks.

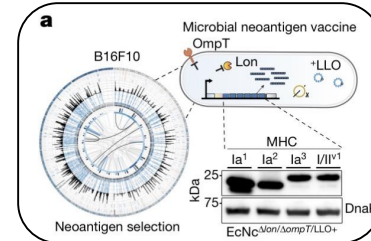


Figure 2: Engineered microbial neoantigen expressing **tumor specific colonization and antigen specific T-cell induction** in B16F10 [1].

Gene of Interest Characteristics [1]

1. **LLO (Listeriolysin O) gene:** A pH dependent pore-forming protein drive from *Listeria* that permeabilizes the phagolysosomal membrane, allowing recombinant neoantigens to escape into the cytosol.
2. **Lon protease gene (Δ lon):** A bacterial protease responsible for degrading of the synthetic neoantigen constructs.
3. **OmpT protease gene (Δ ompT):** An outer membrane protease that degrades complement components, helping bacteria evade immune clearance.
4. **Synthetic Neoantigen Construct gene (NeoAg, with 5-mer GS linkers):** The actual tumor-specific antigen encoded for delivery.
5. **Cryptic plasmid removal (EcN $\rightarrow$ EcNc):** To suppress the copy number of transformed recombinant plasmid. Has no direct immunological function on its own.

[1] Redenti, A., Im, J., Redenti, B., Li, F., Rouanne, M., Sheng, Z., Sun, W., Gurbatri, C. R., Huang, S., Komaranchath, M., Jang, Y., Hahn, J., Ballister, E. R., Vincent, R. L., Vardoshivilli, A., Danino, T., & Arpaia, N. (2024). Probiotic neoantigen delivery vectors for precision cancer immunotherapy. *Nature*, 635(8038), 453–461. <https://doi.org/10.1038/s41586-024-08033-4>

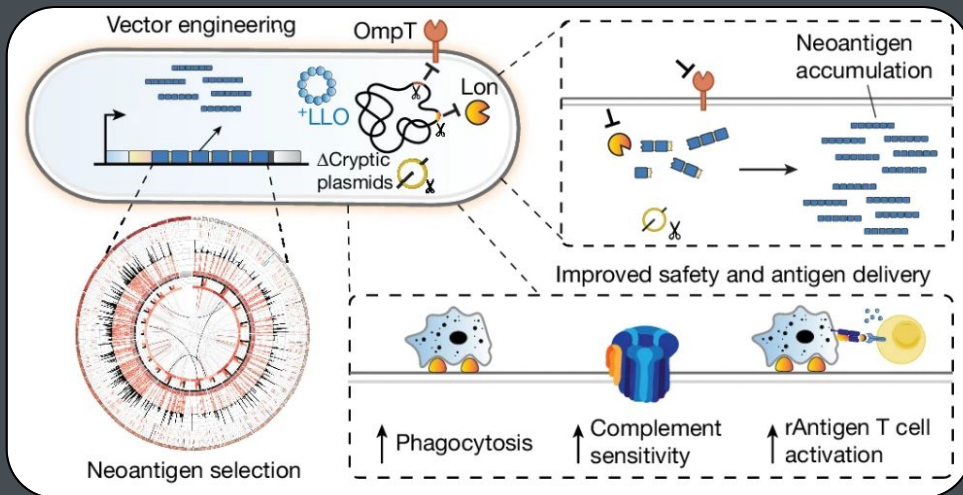


Figure 3: Design of microbial tumour neoantigen vaccines, with a Circos plot of CT26 mutanome [1]

Biological Materials [1]

Engineered probiotic bacteria:

Escherichia coli Nissle 1917 modified to express tumor-specific neoantigen peptides

Cell lines and animal models:

Cultured at 37 °C, 5% CO₂ in DMEM or RPMI-1640 with 10% FBS, GlutaMax, NEAA, and penicillin–streptomycin

Sequenced Exomes: Paired tumour and tail DNA from BALB/c (CT26) and C57BL/6 (B16F10) mice; Exome capture: Agilent SureSelectXT

Sequenced RNA: Tumour RNA extracted via Qiagen RNeasy Mini Kit; libraries built with NEBNext Ultra RNA Prep Kit; mRNA enriched with Oligo(dT) beads

[1] Redenti, A., Im, J., Redenti, B., Li, F., Rouanne, M., Sheng, Z., Sun, W., Gurbatri, C. R., Huang, S., Komaranchath, M., Jang, Y., Hahn, J., Ballister, E. R., Vincent, R. L., Vardoshivilli, A., Danino, T., & Arpaia, N. (2024). Probiotic neoantigen delivery vectors for precision cancer immunotherapy. *Nature*, 635(8038), 453–461. <https://doi.org/10.1038/s41586-024-08033-4>

Methods and Materials

Analytical & Imaging Equipment and Methods[1]

- **Flow cytometer:** Used to analyze immune cell populations and activation (e.g., CD4⁺, CD8⁺ T cells)
- **Fluorescence microscope:** For visualizing cellular interactions and antigen expression
- **Plate reader (ELISA):** To measure cytokine levels such as interferon-gamma
- **Imaging systems / calipers:** For tracking tumor growth and progression in animal models
- **Neoantigen constructs:** Synthetic DNA encoding multiple tumor-specific epitopes, joined by 5-mer Gly-Ser linkers, cloned into a stabilized plasmid under constitutive expression
- **Bacterial Engineering & Vaccine Construct Design:** Final platform (EcNcΔlon/ΔompT/LLO+) validated in vitro for phagocytosis uptake, blood clearance, reduced biofilm formation, and APC activation

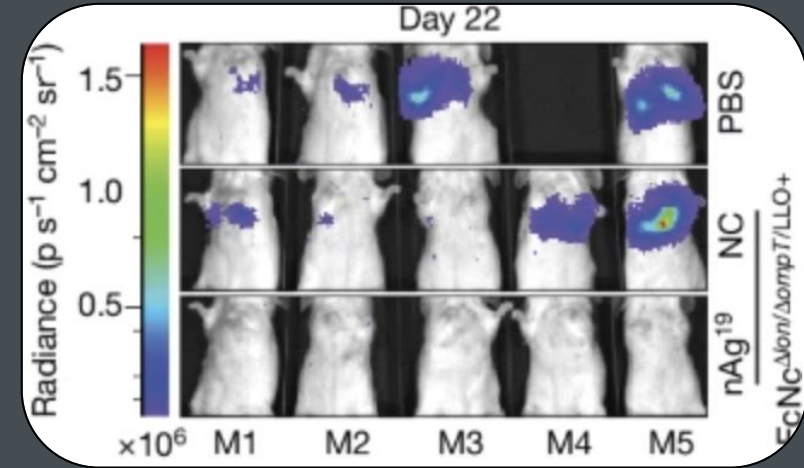


Figure 4: Representative lung metastases luminescence [1]

[1] Redenti, A., Im, J., Redenti, B., Li, F., Rouanne, M., Sheng, Z., Sun, W., Gurbatri, C. R., Huang, S., Komaranchath, M., Jang, Y., Hahn, J., Ballister, E. R., Vincent, R. L., Vardoshivilli, A., Danino, T., & Arpaia, N. (2024). Probiotic neoantigen delivery vectors for precision cancer immunotherapy. *Nature*, 635(8038), 453–461. <https://doi.org/10.1038/s41586-024-08033-4>

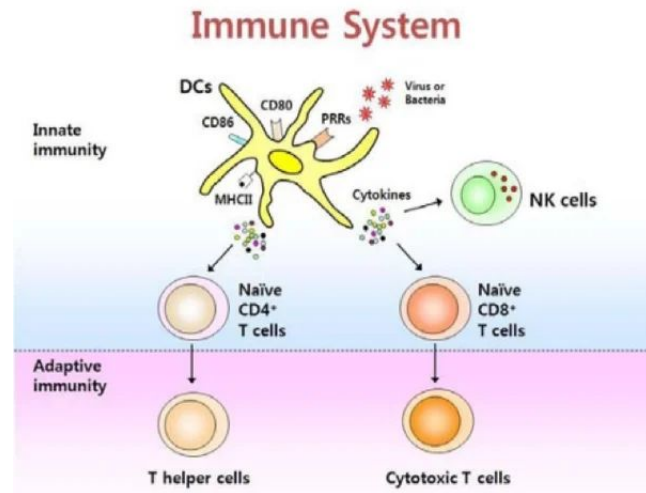
Expected Positive Outcomes

Application Level Positive outcomes:

1. Priming tumor-specific T cells can lead to migration and help destroy other untreated metastases [1, 2]
2. Activation of bystander T cells and Natural Killer cells expands the total anti-tumor attack [2, 5]
3. Prevention of cancer reoccurrence through systematic immunity that recognizes tumor antigens [5]

Gene-Level Positive Outcomes:

1. Integration of LLO allows for delivery of neoantigens directly to the APC cytosol [2]
2. Neoantigen presentation primes immune response to cancer [2]
3. Modification of bacterial surface components increases their susceptibility to be “eaten” by cDC2s and a greater immune response in return [4]



Expected Negative Outcome

Hasna

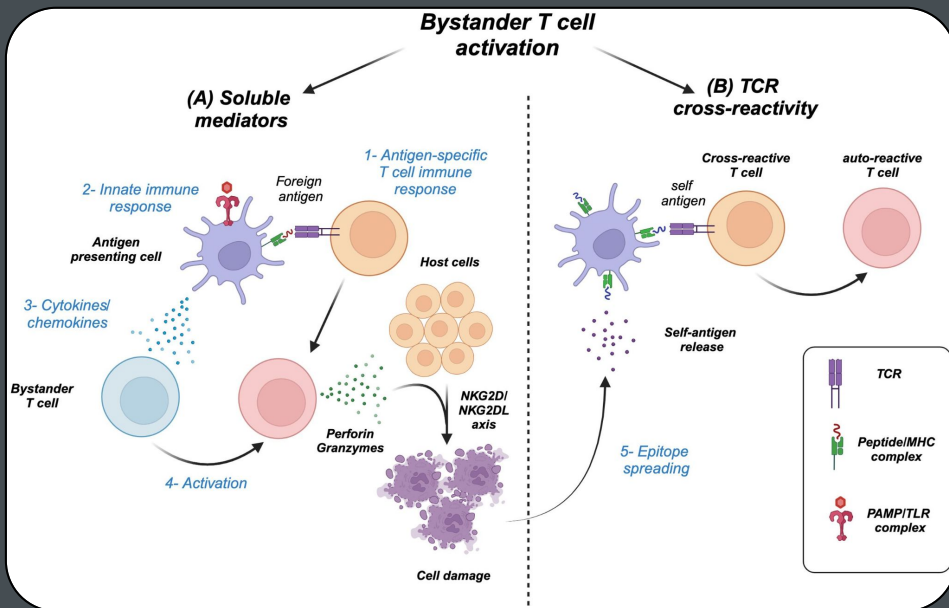


Figure 5: Immune dysregulation leading to bystander activation and auto-reactive T-cells. [5]

Application-Level Negative Outcomes

1. Loss of efficacy if neoantigens are poorly encoded or unstable [1]
2. No therapeutic effect when CD4⁺ or CD8⁺ T-cells are absent [1]
3. Underreactive immunity with weak APC activation and limited tumor control [4]
4. Overreactive immunity with excessive cytokines or inflammation [4]

Gene-Level Negative Outcomes

1. Low neoantigen expression due to plasmid instability [1]
2. Loss of LLO or NeoAg function from mutation or misfolding [3]
3. Other bacterial proteases degrading antigens despite $\Delta lon/\Delta ompT$ deletions [1]

[1] Redenti, A., Im, J., Redenti, B., Li, F., Rouanne, M., Sheng, Z., Sun, W., Gurbatri, C. R., Huang, S., Komaranchath, M., Jang, Y., Hahn, J., Ballister, E. R., Vincent, R. L., Vardoshivili, A., Danino, T., & Arpaia, N. (2024). Probiotic neoantigen delivery vectors for precision cancer immunotherapy. *Nature*, 635(8038), 453–461. <https://doi.org/10.1038/s41586-024-08033-4>

[3] Schumacher, T. N., & Schreiber, R. D. (2015). Neoantigens in cancer immunotherapy. *Science* (New York, N.Y.), 348(6230), 69–74. <https://doi.org/10.1126/science.1264497>

[4] Iwasaki, A., & Medzhitov, R. (2015). Control of adaptive immunity by the innate immune system. *Nature immunology*, 16(4), 343–353. <https://doi.org/10.1038/ni.3123>

[5] Yosri, M., Dokhan, M., Aboagye, E., Al Moussawy, M., & Abdelsamed, H. A. (2024). Mechanisms governing bystander activation of T cells. *Frontiers in Immunology*, 15, 1465889. <https://doi.org/10.3389/fimmu.2024.1465889>

Methods to Measure Outcomes

1. Neoantigen Expression:

ELISA/Immunoblotting can quantify levels of proteins in serum. FLOW Cytometry can measure expression of Neoantigens in isolated PBMCs, specifically Dendritic cells, T Cells and B Cells.

2. T Cell/ B Cell Activity Determination:

Using assays like T Cell/B Cell co-culture we can determine if the body is adapting to neoantigen presentation

3. Mouse Models:

Mouse Models with cancer can be given bacterial neoantigen primer as a pre-treatment. Tumor measurement after 14–21 days could determine changes in mouse tumor clearance/prevention.

Direct ELISA

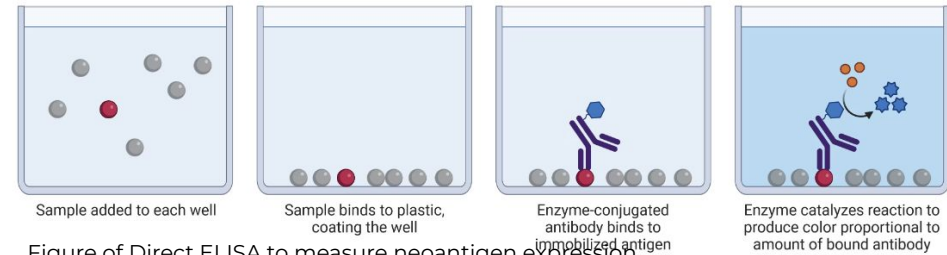


Figure of Direct ELISA to measure neoantigen expression
<https://www.antibodies.com/applications/elisa>

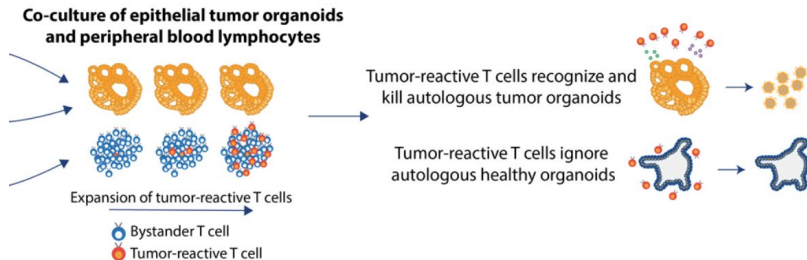
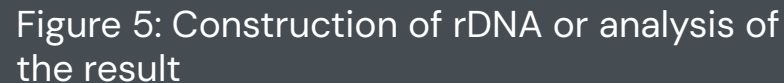


Figure of T Cell co culture example
<https://blog.crownbio.com/how-to-use-tumor-organoids-for-immuno-oncology-applications>

Figure of experimental mouse model example
<https://brainstuff.org/blog/why-do-scientists-study>





1. <https://www.addgene.org/137923/>
2. <https://www.ncbi.nlm.nih.gov/gene/168400>

Replicated a similar protein in snapgene to mimic the tumor targeting EcN.

1. Plasmid #137923 – E Coli. Nissle 1917
2. Listeriolysin O
3. pBAD promoter Plasmid as stabilized plasmid
4. CT26 Mutantome

References

- 1) Redenti, A., Im, J., Redenti, B., Li, F., Rouanne, M., Sheng, Z., Sun, W., Gurbatri, C. R., Huang, S., Komaranchath, M., Jang, Y., Hahn, J., Ballister, E. R., Vincent, R. L., Vardoshivilli, A., Danino, T., & Arpaia, N. (2024). Probiotic neoantigen delivery vectors for precision cancer immunotherapy. *Nature*, 635(8038), 453–461. <https://doi.org/10.1038/s41586-024-08033-4>
- 2) Canale, F. P., Basso, C., Antonini, G., Perotti, M., Li, N., Sokolovska, A., Neumann, J., James, M. J., Geiger, S., Jin, W., Theurillat, J. P., West, K. A., Leventhal, D. S., Lora, J. M., Sallusto, F., & Geiger, R. (2021). Metabolic modulation of tumours with engineered bacteria for immunotherapy. *Nature*, 598(7882), 662–666. <https://doi.org/10.1038/s41586-021-04003-2>
- 3) Schumacher, T. N., & Schreiber, R. D. (2015). Neoantigens in cancer immunotherapy. *Science (New York, N.Y.)*, 348(6230), 69–74. <https://doi.org/10.1126/science.aaa4971>
- 4) Iwasaki, A., & Medzhitov, R. (2015). Control of adaptive immunity by the innate immune system. *Nature immunology*, 16(4), 343–353. <https://doi.org/10.1038/ni.3123>
- 5) Yosri, M., Dokhan, M., Aboagye, E., Al Moussawy, M., & Abdelsamed, H. A. (2024). Mechanisms governing bystander activation of T cells. *Frontiers in Immunology*, 15, 1465889. <https://doi.org/10.3389/fimmu.2024.1465889>